Mo Tu We Th Fr Sa Su Date:

Rail Guard

Catching the bolt Before It Breaks — a vibration-based early-warning system for railway fishplate joints.

Engineer: Mohit Sheth

The Problem: Across India's railway track of approximately 70 000 km in length, workers have to walk along large lengths of track, tapping each bolt in fishplates. This process is slow and inefficient.

Mo Tu We Th Fr Sa Su

Date:

About Me:

My interests: Math, Coding, Robotics and playing Badminton.

My skills: Python coding experience of 6+ years and previous experience in several robotics projects.

Mo Tu We Th Fr Sa Su Date:

Day 1 - Project Introduction | 2nd May 2026

Learnt the difference between:
Idea, Prototype and a Market-ready ProducT.

Learnt feasability checks for ideas based on the following parameters:
- Time
- Resources available
- Budget

Decided on doing an innovation project - solving a real problem.

Domain: Mechanical and Electronics + AI and ML use.

I also started my project Ideation.

Mo Tu We Th Fr Sa Su

Date:

May 2nd Continued

Idea 1:
- Dyslexia-detection tool using eye-tracking
- Sign-language recogniser that reads facial grammar.
- AI Sports Performance Coach (my favorite idea of the three) It uses a vision System and Machine Learning to analyse an athlete's technique in real time.

The above ideas are decent, but don't click yet and also some are too niche or explored.

Ideation

Mo Tu We Th Fr Sa Su Date :

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Day 2 - Hunting Problems | May 9th

Rather than just searching for an idea there are concrete steps that can ~~aid~~ produce better quality ideas.

Literature Review:

- After ~~identyf~~ identifying Problem Statement + Idea search for similar projects to your idea on:
  - Google Scholar
  - Research Gate
  - arXiv
  - Check with professors

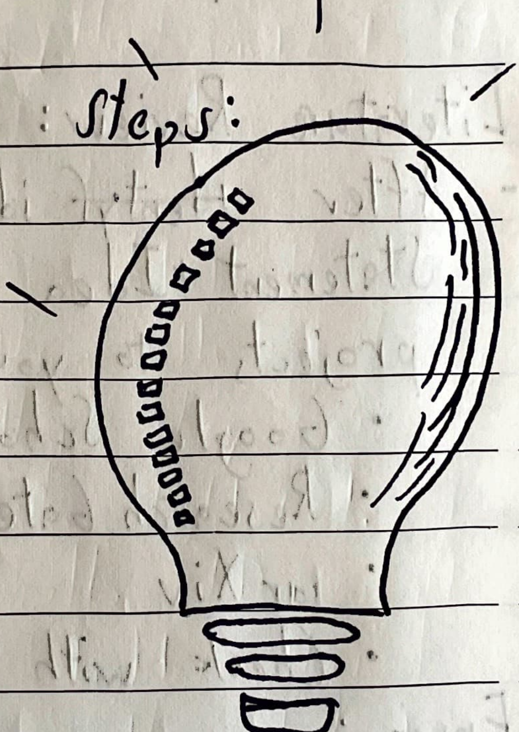
Ensures:

- No project like the idea already exists
- No project solves my problem better than me.
- No research paper proves my assumption wrong.
- Any existing work my project can build on.

Other factors to consider:

- Feasability
- Impact
- Real-world application
- Relation to interest

Project Ideation steps:



Interest
↓
Brainstorming
↓
Problem Statement
↓
Impact of Solution
↓
Literature Review
↓
Feasability
↓
FINER Test
(Feasible, Interesting, Novel, Ethical, Relevant)

Mo Tu We Th Fr Sa Su Date:

Day 3 - Finalising the Idea | May 16

First step: I found 10 unique problem statements and ideas over the last week

Next: I eliminated any idea that did not interest me or did not match the criteria decided on Day 2.

Ended up at 3 ideas of high quality

Idea 1: Underground Robot Network

Problem: Deep Underground Mining creates a communication blackout that makes miners unreachable. Miners cannot notify if they are injured.

Solution: One main robot carries 3 sub-robots inside it.

As the system enters the tunnel and reaches the first point where the signal weakens a sub-robot is deployed and stays behind as a live relay node.

Mo Tu We Th Fr Sa Su   Date:

May 16 continued

Idea 2: Rail Guard

Problem: A fishplate is a steel plate clamping two rail-sections end-to-end. As trains pass, vibration gradually loosens these bolts. A loose fishplate allows the joint to flex leading to rail misalignment and derailment.

Solution: Accelerometers detect the vibration patterns of the bolts, and determine whether they are loose and need to be tightened. Then the loose bolts ID is sent so workers only tighten them rather than walking the whole track.

Mo Tu We Th Fr Sa Su Date:

May 16 continued ✓

Idea 3: Mechanical power presses for smaller companies

Problem: Workers regularly lose fingers and sheet metal feeding generates vast quantities of scrap.

Solution: A camera scans the irregular scrap and generates a precise digital outline. A nested algorithm ~~de~~ computes the optimal layout. A CNC-controlled cutter with automatic sheet movement then cuts all parts without human hands near the tooling.

Mo Tu We Th Fr Sa Su Date:

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Day 4 - Learnt more about Idea | May 23rd

Final Idea: Rail Guard

Most feasable + most interesting

Today, I researched how this project would work - ~~what~~ an MPU 6050 accelerometer would be connected to ~~an~~ a mini ESP-32.

I understood the concept and how ~~we~~ it would work.

Mo Tu We Th Fr Sa Su Date:

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Day 5- First BOM | May 30th

First Bill of Materials

I made a rough outline of my BOM. Some of the components ~~were pricey~~ pushed the total cost up significantly.

Day 6 - Final BOM June 7th

Refined BOM, looked for cheaper alternatives.

The project is now more affordable and simpler to execute.

I ordered the components from Roby.in and amazon.in.

Mo Tu We Th Fr Sa Su

Date:

Day 7 - First Hardware Test | 13th June

Today I brought all my hardware and ~~con~~ wired a mini ESP-32S3 to an MPU 6050 ~~accelerato~~ accelerometer.

Mo Tu We Th Fr Sa Su Date:

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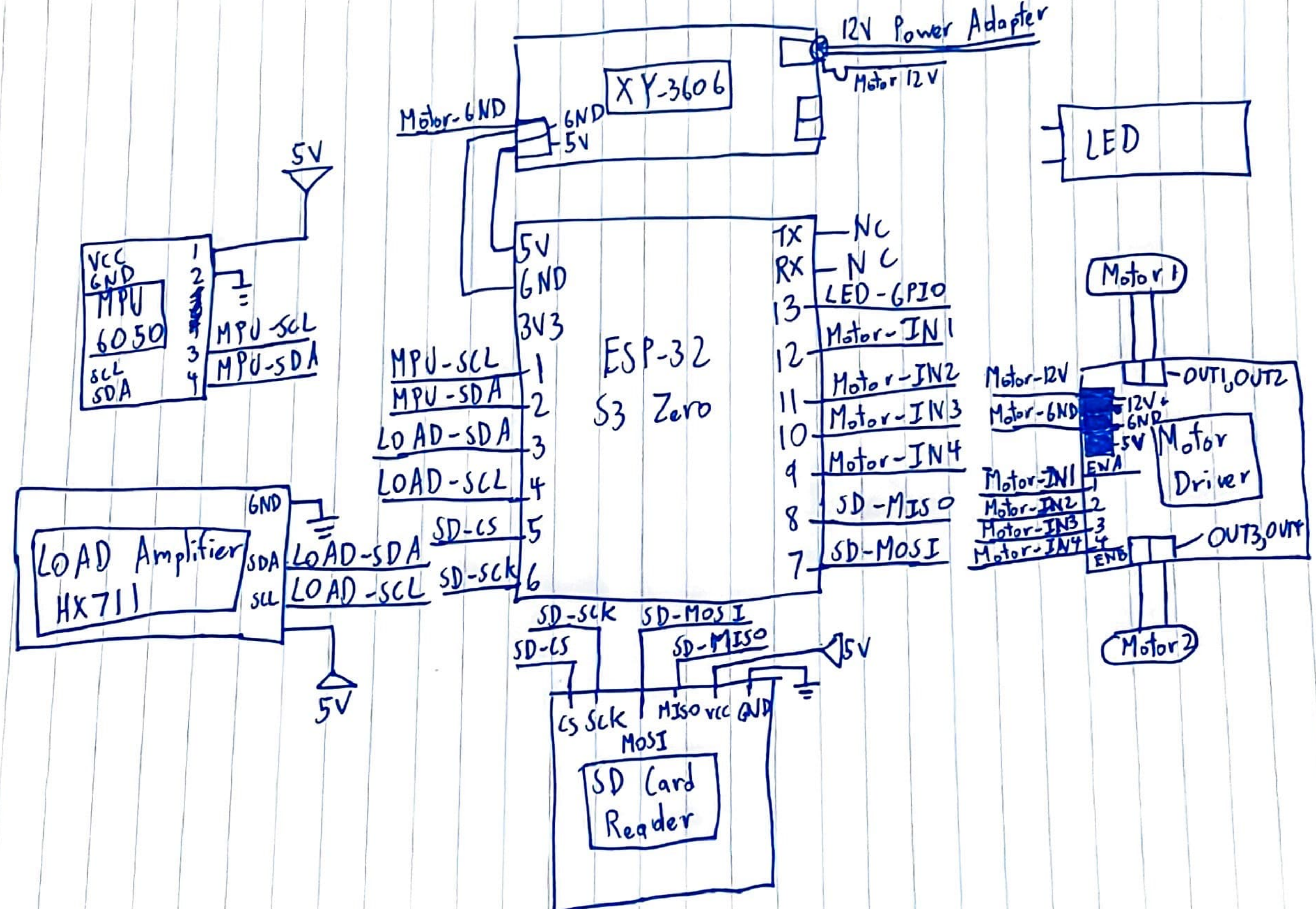
Day 8- Project Schematic | 20th June

Today I made my project schematic.
Steps I followed:
1. Made list of components used
2. Got spec sheet and pinout diagram for each component in a folder
3. Made excel sheet for what each ESP-32 S3 Zero pin is for
4. Made Project Schematic

Day 9: Finished soldering of components onto circuit. | 27th June
Also Finished Wiring between components.


12V Power Adapter
Motor 12V
XY-3606
Motor-GND
GND
5V
LED
5V
VCC
GND
MPU 6050
SCL
SDA
1
2
3
4
MPU-SCL
MPU-SDA
5V
GND
3V3
ESP-32
S3 Zero
TX
RX
NC
NC
13
12
11
10
9
8
7
LED-GPIO
Motor-IN1
Motor-IN2
Motor-IN3
Motor-IN4
SD-MISO
SD-MOSI
MPU-SCL
MPU-SDA
LOAD-SDA
LOAD-SCL
SD-CS
SD-SCK
5
6
Motor 1
Motor-12V
Motor-GND
OUT1,OUT2
12V+
GND
5V
Motor Driver
ENA
ENB
Motor-IN1
Motor-IN2
Motor-IN3
Motor-IN4
OUT3,OUT4
Motor 2
GND
LOAD Amplifier
HX711
SDA
SCL
LOAD-SDA
LOAD-SCL
5V
SD-SCK
SD-MOSI
SD-MISO
SD-CS
5V
CS
SCK
MOSI
MISO
VCC
GND
SD Card Reader